

Go Backend 4Week Mastery Plan (with Roadmap Diagram)

This document provides a structured 4week plan to learn Go for backend development, including REST APIs, authentication, database integration, sessions, and clean architecture. A visual roadmap diagram is included to help you understand the learning progression.

VISUAL ROADMAP DIAGRAM ----- Go Basics (types, slices, maps) | v
JSON Handling | v HTTP (net/http → Gin) | v REST API Design | v Authentication (JWT)
| v Sessions & DB Layer | v Clean Architecture | v Docker + Deployment | v
Microservices Basics

WEEK 1 — Go Fundamentals Goal: Learn only the Go essentials required for backend development. • Variables, types, strings • if, switch, for, for range • slices & maps • functions (multiple returns) • error handling (err != nil) • defer • basic pointers • JSON encode/decode • small exercises with structs

WEEK 2 — Build Your First API (Gin) Goal: Build a real working API with routing, handlers, and simple frontend templates. • Initialize Gin server • /health endpoint • /photos GET/POST (in-memory) • file upload endpoint • middleware basics • serve static images • simple frontend with HTML templates + Bootstrap

WEEK 3 — Authentication + Database Goal: Implement secure registration, login, JWT, refresh tokens, and session tracking. • Connect Postgres (Dockerized) • users table + password hashing (bcrypt) • /auth/register + /auth/login • generate JWT access token + refresh token • sessions table (multi-device login) • refresh token rotation • logout + logout-all endpoints • protected routes (/me)

WEEK 4 — Clean Architecture + Deployment Goal: Turn your project into a production-ready backend service. • Folder structure: cmd/, internal/http/, internal/auth/, internal/persistence/ • interfaces for repositories and services • unit tests for auth logic • Dockerfile + docker-compose • environment configs (.env) • HTMX-based frontend pages • logging + error formatting • final polish + documentation

NOTES

(Use this section for personal notes.)

